

FORMULARIUM EXAMEN PRODUCTIETECHNIEKEN & -SYSTEMEN

$$T_e = \left(\frac{1}{m} - 1 \right) \frac{K_G}{K_M + K_L} \quad \text{of} \quad T_e = -v_1 \frac{C}{A} \qquad T_p = \left(\frac{1}{m} - 1 \right) \cdot T_{cT}$$

$$C_T = \frac{C_{TVB} \cdot VB^n}{h^p \cdot b^q} \quad (q \approx 0)$$

$$VB = A \cdot t^\beta$$

$$D_{0,2} = K \cdot N^{-\mu}$$

$$v_{60} = C_v A^{-\varepsilon_v}$$

$$\text{Parabolische beitelvorm: } y = \frac{x^2}{2 \cdot r_g}$$

$$\Delta l = l \cdot \alpha \cdot \Delta T$$

$$E_{\text{staal}} = 210 \text{ GPa} \qquad \alpha_{\text{staal}} = 12 \cdot 10^{-6} \text{ } 1/^{\circ}\text{C}$$

$$z = 0.2$$

$$\sigma_b = \frac{M_b}{I} y \qquad I = \frac{b \cdot h^3}{12}$$

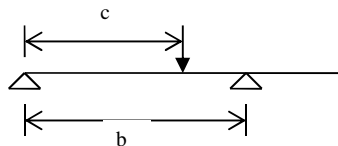
$$I_o = \frac{\pi \cdot d^4}{64}$$

$$M = \frac{60 \cdot P}{2 \cdot \pi \cdot n}$$

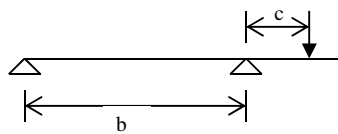
$$F_b = C_b \frac{s^2 \cdot b \cdot R_m}{w}$$

$$F_c = 2.5 \cdot F_p$$

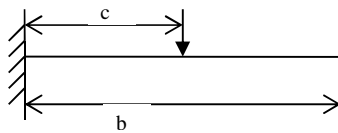
$$h(\varphi) = f_z \cdot \sin \varphi \cdot \cos \lambda \cdot \sin \kappa$$



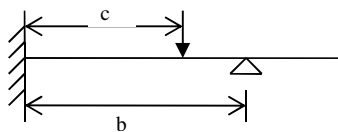
$$y_F = \frac{c^2(b-c)^2}{3 \cdot E \cdot I \cdot b} F \qquad \text{Zo } c = \frac{b}{2} \qquad y_F = \frac{b^3}{48 \cdot E \cdot I} F$$



$$y_F = \frac{c^2(b+c)}{3 \cdot E \cdot I} F$$



$$y_F = \frac{c^3}{3 \cdot E \cdot I} F$$



$$y_F = \frac{(b-c)^2 c^3 (4b-c)}{12 \cdot E \cdot I \cdot b^3} F$$

Slijpen

$$Q_w = a_e \cdot a_p \cdot v_w$$

$$Q'_w = \frac{Q_w}{b_{s\text{ eff}}}$$

$$h_{eq} = \frac{Q'_w}{v_s}$$

$$l_c = \sqrt{2 \cdot a \cdot r_s}$$

Omvormen

$$\sigma = K \varepsilon^n$$

$$\dot{\sigma} = C^* \varepsilon^m$$

$$\bar{Y}_f = \frac{K \varepsilon^n}{n+1}$$

$$\varepsilon = \ln \frac{h_0}{h_f}$$

$$\dot{\varepsilon} = \frac{\varepsilon}{L/v_r}$$

$$L = \sqrt{R(h_0 - h_f)} = \sqrt{R\bar{d}}$$

$$d_{max} = \mu^2 R$$

$$\theta_{max} = \tan^{-1}(\mu)$$

$$F = wL \left(\frac{2}{\sqrt{3}} \bar{Y}_f \right) \left(1 + \frac{\mu L}{2h_{av}} \right) \quad \text{met} \quad h_{av} = \frac{h_0 + h_f}{2}$$

$$T = \frac{FL}{2}$$

$$P = 2\omega T$$

$$A_b = \alpha(r_b + C_a \cdot s_0)$$

$$\alpha_t = 3 \cdot \frac{R_e}{E} \cdot \frac{r_b + C_a \cdot s_0}{s_0} \cdot \alpha_b$$

$$F_b = C_b \cdot \frac{b \cdot s_0^2 \cdot R_m}{w} \quad \text{of} \quad F_b = C_b \cdot b \cdot s_0 \cdot R_m$$

Scheiden

$$u = a \cdot t \qquad F_{th} = \tau \cdot L \cdot t \qquad \text{met} \qquad \tau = 0.75 \cdot R_m$$

Oervormen

$$H = \rho V \{ c_s (T_m - T_0) + H_f + c_l (T_p - T_m) \}$$

$$Q = A_1 \cdot v_1 = A_2 \cdot v_2$$

$$v_2 = \sqrt{2 \cdot g \cdot h_1}$$

$$T_{MF} = \frac{V}{Q}$$

$$T_{TS} = C_m \left(\frac{V}{A} \right)^n \quad \text{met} \quad n = 2$$

$$q = h_{\text{eff}}(T - T_0)$$

$$h_c = \frac{\kappa_s}{X}$$

$$h_R = \frac{\sigma(\varepsilon T^4 - \alpha T_0^4)}{T - T_0}$$

$$\text{Parallel: } h_{\text{eff}} = \sum_i h_i \qquad \text{Series: } h_{\text{eff}} = \left[\sum_i \frac{1}{h_i} \right]^{-1}$$